A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 - Generated practice paper Set 22

## Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion<br>– waste products of one process are raw materials of another so wastes do not build up<br>**1** Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere<br>**1** Allowing more plant growth, further stabilising climate and soils<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **transpiration** → water lost as vapour from the leaves | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.2** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** Lack of ancient historical / instrumental data<br>**1** Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions<br>**1** Lack of sophisticated equipment / inability to measure many factors<br>**1** Limited coordination between researchers<br>– lack of data collection in many areas (oceans, poles)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **reliability** → repeatable / gives consistent results when repeated<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument | 3 |
| **01.3** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell<br>**1** Air bubbles trapped in ice give the composition of the past atmosphere ($CO_2$, methane)<br>**1** Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments)<br>**1** More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3. | |
| **02.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** The new opportunity is overwhelmingly attractive (economic benefit)<br>**1** Lack of knowledge / understanding of the impacts<br>**1** The impacts occur elsewhere or much later, so are not evident to those who benefit<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Cultural belief that resources should be exploited to satisfy needs only<br>**1** Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **02.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Resources are brought in from other areas<br>**1** If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **02.4** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Easier access to energy has made many new activities possible<br>**1** The scale of activities has greatly increased<br>**1** Toxic wastes are produced that cannot be decontaminated by natural processes<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1. | |
| **03.1** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish<br>**1** Species occupy different niches / trophic levels so they do not compete and all food sources are used<br>**1** Bottom feeders re-suspend nutrients boosting plankton / plant growth<br>**1** Predators control over-breeding species (tilapia) so remaining fish grow larger<br>– total yield exceeds any single species<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.2** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1 IMTA:** fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton)<br>**1** Wastes of one species become nutrients for another so less pollution and extra harvests<br>**1 Aquaponics:** aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress)<br>**1** Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish)<br>– rice-fish paddies similar<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **aquaculture** → fish or shellfish farming<br>• **deoxygenation** → loss of dissolved oxygen from the water<br>**Do not credit**<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts | 4 |
| **03.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages<br>**1** Production still depends on (and removes juveniles from) the wild population<br>– carnivores need large fish inputs<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **aquaculture** → fish or shellfish farming<br>**Do not credit**<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3. | |
| **04.1** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses<br>**1** Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population<br>**1** Deep-water aggregations easily located and caught by trawling<br>**1** Recovery takes decades so fisheries have had to be closed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **04.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young<br>**1** Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 2 |
| **04.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort<br>**1** Masks the decline of the breeding population until a sudden collapse<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **04.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Neither is a target species but both are caught in demersal trawls<br>– skate now extinct in the Baltic and critically endangered elsewhere<br>**1** Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2. | |
| **05.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** The range of an environmental factor within which a species / population can survive (survival highest at the optimum, falling towards the limits)<br>**1** X survives from about pH 5.5 to 8.5 (wide range)<br>– Y only from about pH 6.0 to 8.0 (narrow range)<br>**1** Both have the same optimum (pH 7.0)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **range of tolerance** → the range of conditions in which a species can survive | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **05.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** X: 60% survive at pH 6.0 whereas only 5% of Y survive<br>**1** X has a larger gene pool / wider range of tolerance so some individuals are adapted to the new conditions<br>**1** Survivors breed and their offspring are adapted so the population rebuilds<br>– Y is likely to die out locally<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **gene pool** → the range of alleles / genes present in a population<br>• **range of tolerance** → the range of conditions in which a species can survive | 3 |
| **05.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Reveals the habitat requirements / which abiotic factor is near the edge of tolerance<br>**1** Management can make small beneficial changes to that factor (eg liming, shading, water levels) to keep conditions within the range and increase survival<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **range of tolerance** → the range of conditions in which a species can survive<br>• **abiotic** → non-living / physical factors | 2 |
| **05.4** | **Any 2 of 5**<br>1 mark each for any 2 of the 5 points below, in any order, maximum 2.<br>**1** Food supply<br>**1** Pollinators<br>**1** Seed dispersal species<br>**1** Disease / pathogens<br>**1** Predators / competitors<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **biotic** → living factors / effects of other organisms<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-73; AQA spec 3.1.3.1. | |
| **06.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1 A:** The readings are precise but may not be accurate<br>**Also accept**<br>• **precise** → readings close together / fine resolution<br>• **accurate** → close to the true value | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **06.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** A result that differs from other results with which it was expected to be similar<br>**1 Repeat the study / measurement:** if it recurs it is a 'real' result caused by another variable<br>– if not it was an error and can be excluded<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **anomalous** → an outlier / a result that does not fit the pattern<br>**Do not credit**<br>• discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty | 2 |
| **06.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** To test the reliability of the method / equipment<br>**1** To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **reliability** → repeatable / gives consistent results when repeated<br>**Do not credit**<br>• 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1. | |
| **07.1** | **1 mark each**<br>1 mark for each of the 5 points below.<br>**1 Timber:** building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood)<br>**1 Fuel:** fuelwood and charcoal - still the main energy source for most people in LEDCs<br>**1 Food:** fruit, nuts, bushmeat, livestock fodder<br>– crops of forest origin (coffee, cacao, bananas, Brazil nuts)<br>– pigs and chickens were forest animals<br>**1 Fibres:** paper (wood pulp cellulose), cotton, viscose / rayon<br>**1 Medicines:** quinine (cinchona), taxol (yew), aspirin (willow)<br>– most species not yet researched<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 5 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time<br>– total forest carbon (vegetation plus soil) is about double that in the atmosphere<br>**1** Growing trees sequester CO2 by photosynthesis and release oxygen<br>– forest soils hold more carbon than the trees (especially boreal)<br>**1** Planting / protecting forests removes CO2 and reduces the enhanced greenhouse effect<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime<br>• **enhanced greenhouse effect** → extra warming caused by added greenhouse gases<br>**Do not credit**<br>• 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem<br>• a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation | 3 |
| **07.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Highest transpiration of any ecosystem increases downwind rainfall<br>– interception increases evaporation<br>– deep organic soils retain water and delay runoff so river flow is more even and flooding reduced<br>**1** Low albedo absorbs sunlight<br>– heat stored and released at night reduces day-night temperature extremes<br>– canopy microclimate: lower light, wind, higher humidity<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **transpiration** → water lost as vapour from the leaves<br>• **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back<br>**Do not credit**<br>• 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1. | |
| **08.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Images / maps of large areas<br>– land-use change, deforestation, flooding<br>**1** Vegetation density and surface temperature<br>**1** Wind velocity, sea-surface altitude, wave height, oil pollution<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **08.2** | 1 mark for the point below.<br>**1** Microwaves pass through clouds (unlike visible light)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| **08.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected<br>**1** No need for researchers to visit inaccessible areas, reducing cost and disturbance<br>– data are standardised across the whole area<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.<br>• 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss | 2 |
| **08.4** | **Any 4 of 5**<br>1 mark each for any 4 of the 5 points below, in any order, maximum 4.<br>**1** Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night<br>**1** Each tiger has a unique pattern of facial / body stripes<br>**1** Images are compared with the database so individuals are recognised rather than counted twice<br>**1** The number of different individuals recorded gives a minimum population<br>– repeated captures give territory size, movements, lifespans and social groupings<br>**1 Limitation:** only works for species with unique visible features<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **infrared** → heat radiation / long-wave radiation re-emitted by the Earth | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6. | |
| **09.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Upwelling currents bring nutrients from the seabed to surface waters<br>**1** 24-hour daylight in summer allows a huge phytoplankton bloom<br>**1** Algae also grow under sea ice where krill shelter and feed<br>– krill support most of the food web<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **09.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Krill is the main food of baleen whales, penguins, seals and fish so over-harvesting removes food from most of the web<br>**1** Toothfish overfishing removes the food of sperm whales and elephant seals<br>**1** Long-lining for toothfish catches and drowns albatrosses on the hooks (by-catch)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 3 |
| **09.3** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Ice on land stores 70% of the Earth's fresh water, keeping sea levels low<br>**1** High albedo of ice reflects sunlight, limiting warming (reduction in ice area causes further warming)<br>**1 Carbon sequestration:** carbon in algae eaten by krill sinks to the seabed in krill faeces<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back<br>**Do not credit**<br>• 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed | 3 |
| **09.4** | 1 mark for the point below.<br>**1** Stronger winds spreading ice<br>– more snowfall from warmer, moister air freezing onto slush<br>– fresh meltwater from land forming a surface layer that freezes more easily (long-term trend still expected to be a decline)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 67-71; AQA spec 3.1.2.3.4. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **10.1** | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **In-situ:** species remain in natural conditions with the whole community and inter-species relationships<br>– protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers)<br>• **Limits of in-situ:** external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations<br>• **Ex-situ:** captive breeding when in-situ is impossible or insufficient<br>– selection criteria<br>– difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation)<br>– techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning)<br>– seed banks<br>– successes (Hawaiian goose, Iberian lynx, red kite, large blue)<br>• **Limits of ex-situ:** expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring)<br>• **Judgement:** ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to<br>– both needed, in-situ fundamental<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content.<br>• **abiotic** → non-living / physical factors<br>• **biotic** → living factors / effects of other organisms | 9 |
| | • **plagioclimax** → a community held back from its climax by human activity<br>**Do not credit**<br>• captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well | |
| | **Model answer - Level 3 (8 marks)** | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | In-situ conservation keeps a species in its natural habitat with the whole community it depends on - food species, pollinators, predators and the abiotic conditions it is adapted to. It is delivered through protected areas such as SSSIs, NNRs, SACs, SPAs, Ramsar sites and Marine Conservation Zones, land ownership by conservation bodies, agri-environment schemes and habitat creation and management. Its weaknesses are that external threats such as pollution and climate change cross reserve boundaries, enforcement and cost, conflicts with other land uses, and the fact that many populations are already small and fragmented. | |
| | Ex-situ captive breeding is used when in-situ protection is impossible or insufficient. It is hard: animals need enough space, the right food, social groupings and breeding triggers, and small founder populations risk inbreeding and hybridisation, which studbooks, artificial insemination and embryo transfer help to manage. It is expensive, works for few species, and captive animals may lose wild behaviour; release succeeds only where suitable habitat exists and with soft release and post-release monitoring. | |
| | The two are not alternatives. Ex-situ methods are a temporary safeguard that can rescue a species from extinction, as with the Arabian oryx, but they only succeed if there is protected habitat to return to. In-situ conservation is therefore fundamental and ex-situ a support for it; the relative importance depends on how close to extinction the species is. | |
| | **Weaker answer - Level 2 (5 marks)** | |
| | In-situ conservation means protecting species in their natural habitat, for example in nature reserves and national parks. This is good because the species stays in the wild with the other species it needs. Ex-situ conservation means captive breeding in zoos and then releasing the animals back into the wild. This can save species that are nearly extinct but it is expensive and animals may not survive when released. Both methods are important and are often used together. | |
| | **What makes the difference** | |
| | • Level 2 defines both approaches correctly but Level 3 gives specific mechanisms (types of protected area, breeding difficulties, release methods) and named examples. | |
| | • Level 3 weighs the limitations of each approach explicitly; Level 2 gives one limitation for ex-situ only. | |
| | • 'Both are important' is not a judgement about *relative* importance; Level 3 explains why in-situ is fundamental and when ex-situ matters most. | |
| **10.2** | 1 mark for the point below.<br>**1** Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.1 | **Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Ecological features:** cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web<br>– specific abiotic requirements (temperature, light, salinity, low turbidity)<br>• **Importance:** quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism<br>• **Vulnerability:** narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching<br>– slow growth so slow recovery<br>– physical damage<br>– overfishing and food-web effects (crown of thorns)<br>– introduced species<br>– dependence on mangroves and seagrass<br>• **Conservation:** MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES<br>• Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant<br>• Evaluation of local vs global threats and justified conclusion | 25 |
| | *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4. | |

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| Question | Marking guidance | Marks |
|---|---|---|
| 11.2 | **Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Need:** population possibly 12 billion<br>– humans raise carrying capacity<br>– trophic level of diet<br>• **Impacts to be reduced:** habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion<br>• **Social, economic and political influences:** consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes<br>• **Strategies:** cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention<br>• **Limits:** little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation<br>• Justified conclusion<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5. | 25 |

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